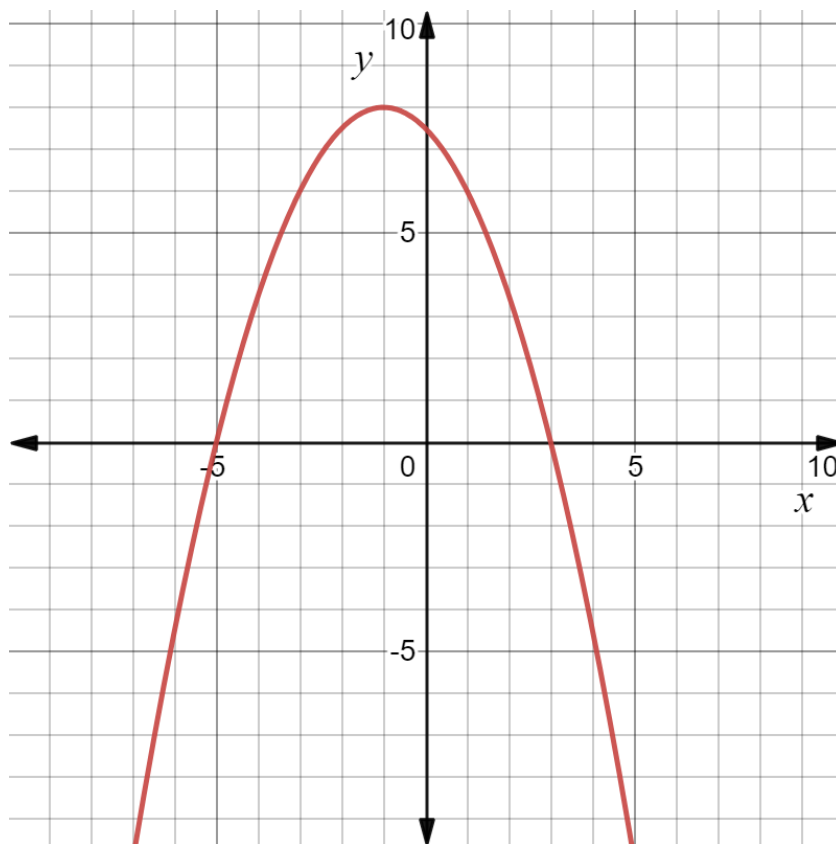


Forms of Quadratic Functions	$y = ax^2 + bx + c$	$y = a(x - h)^2 + k$	$y = a(x - p)(x - q)$
-------------------------------------	---------------------	----------------------	-----------------------

1. An equation is shown.

$$y = -0.5(x + 5)(x - 3)$$

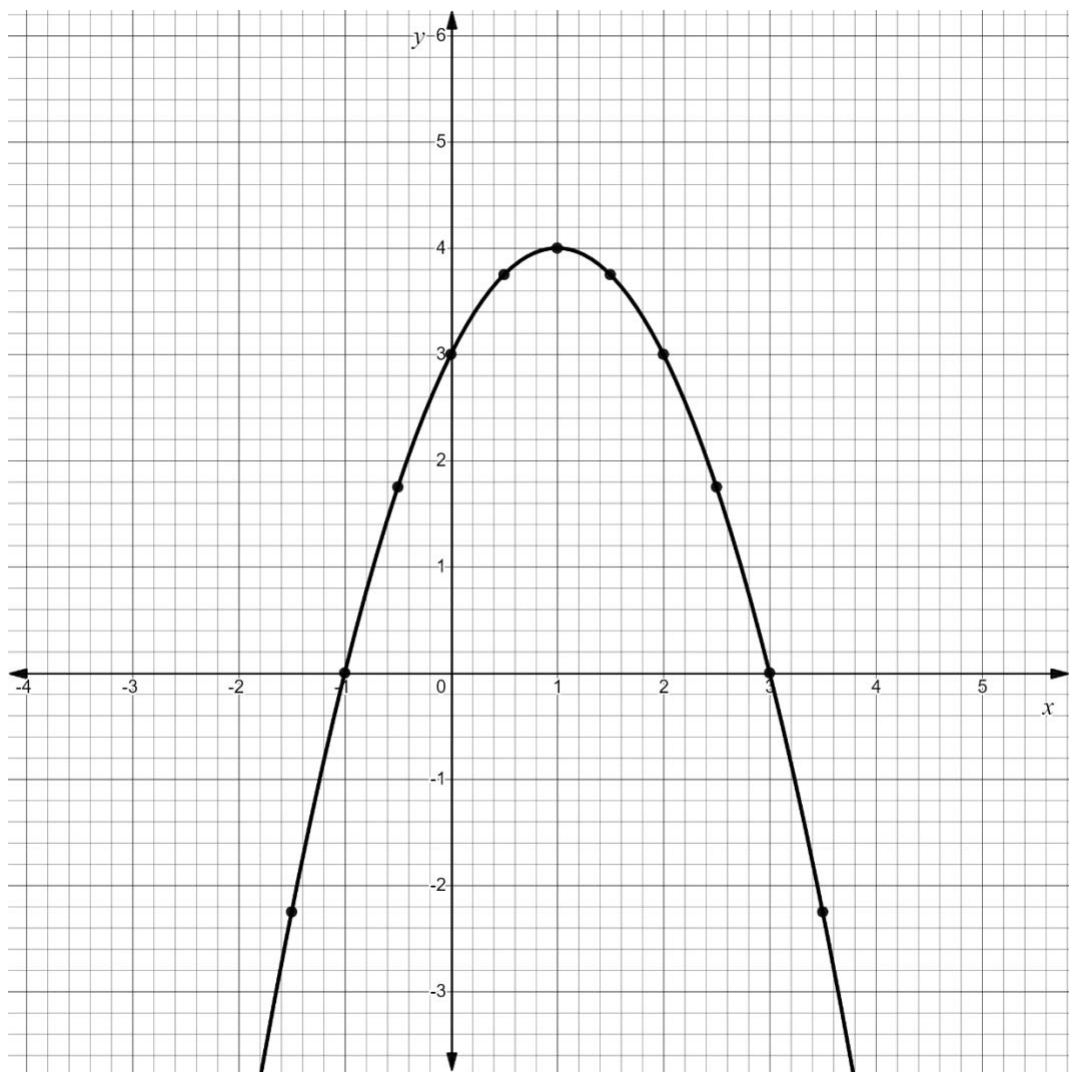
Use the coordinate grid to graph the equation.



2. Rewrite the equation $y = -0.5(x + 5)(x - 3)$ in standard form.

work	
Standard Form	$y = -0.5x^2 - x + 7.5$

3. A quadratic function is graphed on the coordinate grid.



Complete the table.

Name	Key Feature	Name	Key Feature
x-intercepts	$(-1, 0)$ and $(3, 0)$	increasing	$x < 1$
y-intercept	$(0, 3)$	decreasing	$x > 1$
vertex	$(1, 4)$	positive	$-1 < x < 3$
axis of symmetry	$x = 1$	negative	$x < -1$ and $x > 3$
domain	R or $-\infty < x < \infty$ or $\{x \mid -\infty < x < \infty\}$	range	$y < 4$ or $\{y \mid y < 4\}$

4. Rewrite the equation $y = 4x^2 + 40x + 89$ in vertex form.

work	
Vertex Form	$y = 4(x + 5)^2 - 11$

5. Write the quadratic equation $y = x^2 - 6x - 16$ in factored form.

work	
Factored Form	$y = (x + 2)(x - 8)$

6. A table of values for a quadratic function is shown.

x	1	6	8	10	15	17
y	251	31	-1	-1	139	251

Complete each statement.

As $x \rightarrow -\infty$, $y \rightarrow \infty$.

As $x \rightarrow \infty$, $y \rightarrow \infty$.

7. A researcher found that for the years 2013 to 2019, the equation, $y = -0.4(x - 3)^2 + 42$, models the average gas mileage of new vehicles sold in Switzerland, where x is the number of years since 2013 and y is the average gas mileage, in miles per gallon (mpg).

During what year was the average gas mileage for new vehicles sold in Switzerland the greatest?

Year	2016
-------------	------

8. A mathematician used data of the average gas mileage of new vehicles sold in Greece for the years 2010 to 2019 to create a quadratic function that models the data, where x is the number of years since 2010 and y is the average gas mileage, in miles per gallon (mpg). Some of the values are shown in the table of values.

x	0	3	5	7	10	14.574
y	40	52.6	55	52.6	40	0

During what years was the average gas mileage for new vehicles sold in Greece decreasing?

Years	2015 to 2019
--------------	---------------------

9. A social scientist researches the prevalence of undernourishment, as a share of the population. Undernourishment is the main hunger indicator used by the United Nation's Food and Agriculture Organization. The social scientist measures the share of the population which has a caloric (dietary energy) intake which is insufficient to meet the minimum energy requirements defined as necessary for a given population. The Pacific Ocean is home for several small countries such as Samoa, Tonga, and Fiji.

The social scientist found the equation, $y = 0.2(x + 1)(x - 14)$, models the relative change as a percentage, y , of the undernourished population of the small countries in the Pacific Ocean x years since 2001.

- a. Determine the x -intercepts of the equation $y = 0.2(x + 1)(x - 14)$.

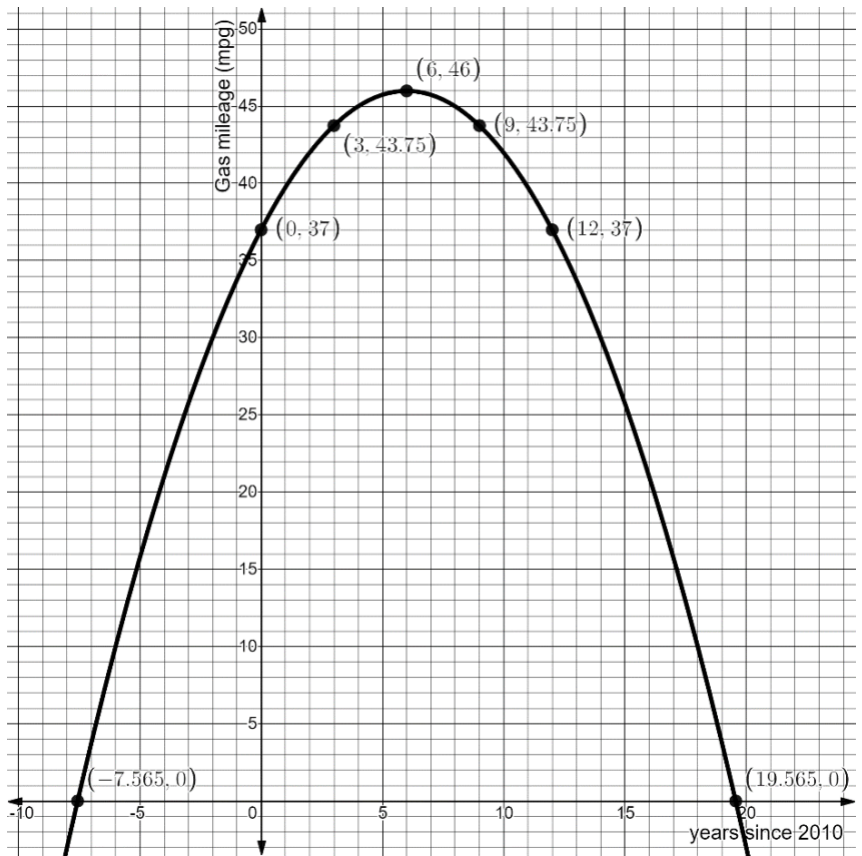
work	
x-intercepts	$(-1, 0)$ and $(14, 0)$

- b. Interpret the x -intercepts of the equation $y = 0.2(x + 1)(x - 14)$ for the given context.

Interpretation of the x-intercepts	The year in which there was no relative change of the undernourished population as a percentage of the small countries in the Pacific Ocean is year 2015. The x-intercepts in this equation are -1 year and 14 years after the year 2001, or years 2000 and 2015, that the relative change was zero. But -1 years after 2001, or year 2000, does not lie in the domain so it doesn't make sense in the context.
--	--

10. The graph of the function, $G(x) = -\frac{1}{4}x^2 + 3x + 37$, models the average gas mileage of new vehicles sold in Sweden for the years 2010 to 2019, where x is the number of years since 2010 and G is the average gas mileage, in miles per gallon (mpg), is shown.

Complete the table by naming the key feature, if any, the coordinate represents and by determining if the coordinate makes sense for the context. Justify your answer.



	Coordinate	Key Feature (write none if it is not a key feature)	Does the coordinate make sense for the context?	Justification
a.	$(-7.565, 0)$	x -intercept	☐ No ☐ Yes	The domain is 0 to 9. The coordinate is not in the domain.
b.	$(0, 37)$	y -intercept	☐ No ☒ Yes	This is the gas mileage for vehicles sold in Sweden in the year 2010.
c.	$(6, 46)$	vertex	☐ No ☒ Yes	6 years after 2010, or year 2016, the average gas mileage for vehicles sold in Sweden is 46 mpg.

- d. Determine the reasonable constraints for the context. Write the constraints in set-builder notation.

Constraints	$0 \leq x \leq 9$ and $37 \leq y \leq 46$
-------------	---